

Abstract Algebra

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1. Sets

1.1. RELATIONS

1.1.1. Set Theory

Definition 1.1 (Tuple)

A tuple (or sometimes **list**) is a finite, ordered collection of elements, possibly with repetitions.

An n -tuple is of length n . A 2-tuple is called an **ordered pair**.

1.1.2. Morphisms

Definition 1.2 (Map)

Subsets of the Cartesian product $A \times B$ are called **relations**. A **map** is a **well-defined** relation written

$$f \subseteq A \times B \quad \text{or} \quad f : A \rightarrow B \quad \text{or} \quad A \overset{f}{\rightarrow} B$$

and is

- **surjective** or **onto** if $f(A) = B$, which is equivalent to being **right invertible** with $f \circ f^{-1} = \text{id}_B$.
- **injective** or **one-to-one** if $a_1 \neq a_2 \implies f(a_1) \neq f(a_2)$, which is equivalent to being **left invertible** with $f^{-1} \circ f = \text{id}_A$.
- **bijective** or a **one-to-one correspondence** if both above apply, which is equivalent to being **invertible**.

Definition 1.3 (Homomorphism)

A **homomorphism** is a map between algebraic structures that preserves the structure.

A group homomorphism $\varphi : G \rightarrow H$ satisfies

$$\varphi(g * \gamma) = \varphi(g) \cdot \varphi(\gamma)$$

for all $g, \gamma \in G$.

Theorem 1.1

If $\varphi : G \rightarrow H$ is a group homomorphism,

- (i) $\varphi(\text{id}_G) = \text{id}_H$
- (ii) $\varphi(g^{-1}) = \varphi(g)^{-1}$ for all $g \in G$

Definition 1.4 (Isomorphism)

An **isomorphism** is a bijective homomorphism.

If there exists an isomorphism between two algebraic structures, they are said to be **isomorphic**, denoted with $\cong$.

Definition 1.5 (Endomorphism)

An **endomorphism** is a homomorphism from a mathematical object to itself.

Definition 1.6 (Automorphism)

An **automorphism** is simultaneously an isomorphism and an endomorphism: a bijective homomorphism from an object to itself.

Example. The **identity map** id_X is a trivial automorphism defined

$$\text{id}_X(x) = x \quad \forall x \in X.$$

Example. An automorphism of a set is called a **permutation**.

1.1.3. Equivalence

Definition 1.7 (Equivalence Relation)

An **equivalence relation** on a set A is a relation $R \subseteq A \times A$ with the following properties:

- (i) **reflexive** $(a, a) \in R \quad \forall a \in A$
- (ii) **symmetric** $(a, b) \in R \implies (b, a) \in R$
- (iii) **transitive** $(a, b), (b, c) \in R \implies (a, c) \in R$

If $(x, y) \in R$ we write xRy or $x \sim y$.

Let R be an equivalence relation on a set A .

Definition 1.8 (Equivalence Class)

The **equivalence class** of an element $a \in A$

$$[a] = \bar{a} = \{b \in A \mid b \sim a\}$$

is the set of all elements related to it.

Example. The modulo m congruence class $[a] = \{a + km \mid k \in \mathbb{Z}\}$.

Definition 1.9 (Quotient Set)

The **quotient set** of A by R

$$A/R = \{[a] \mid a \in A\}$$

is the set of all equivalence classes induced by the relation.

Definition 1.10 (Partition)

A **partition** $\mathcal{P}$ of a set A is a collection of non-empty, disjoint subsets of A whose union is A .

Theorem 1.2

The quotient set A/R forms a partition of A .

Definition 1.11 (Quotient Map)

A **quotient map** is a map $\pi : A \rightarrow A/R$ defined by

$$\pi(a) = [a].$$

Theorem 1.3

Quotient maps are surjective.

1.2. $\mathbb{Z}$

1.2.1. Linear Diophantine Equations

Let a and b be integers with $\text{gcd}(a, b) = d$.

Lemma (Bézout's Identity)

There exist integers x and y such that $ax + by = d$.

Lemma (Euclid's)

If p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.

Theorem 1.4 (Solvability Condition)

The linear Diophantine equation $ax + by = c$ has (infinitely many) solutions if and only if $d \mid c$.

Theorem 1.5 (Solutions)

The solutions, if they exist, are

$$x = x_0 + t \frac{b}{d}, \quad y = y_0 - t \frac{a}{d}, \quad \text{for } t \in \mathbb{Z}.$$

Theorem 1.6 (Euclidean Algorithm)

Let a and b be non-negative integers with $a \geq b$. Then there exist unique integers q and r such that

$$a = qb + r, \quad 0 \leq r < b.$$

Example. We apply the algorithm to compute $\text{gcd}(252, 105)$:

$$252 = 2 \cdot 105 + 42$$

$$105 = 2 \cdot 42 + 21$$

$$42 = 2 \cdot 21 + 0$$

Hence, $\text{gcd}(252, 105) = 21$.

Example. We then use our results to solve $252x + 105y = 21$.

Since $\text{gcd}(252, 105) = 21$, we now work backwards:

$$21 = 105 - 2 \cdot 42$$

$$= 105 - 2 \cdot (252 - 2 \cdot 105)$$

$$= 105 - 2 \cdot 252 + 4 \cdot 105$$

$$= 5 \cdot 105 - 2 \cdot 252$$

$$= -2 \cdot 252 + 5 \cdot 105$$

So a particular solution is $x_0 = -2, y_0 = 5$. The general solutions are

$$x = -2 + 5t, \quad y = 5 - 12t, \quad \text{for } t \in \mathbb{Z}.$$

2. Groups

2.1. DEFINITION

Definition 2.1 (Group)

A **group** $(G, *)$ is a nonempty set G with a binary operation

$$* : G \times G \rightarrow G$$

such that for all $a, b, c \in G$, it satisfies

- (i) (Closure) $a * b \in G$
- (ii) (Associativity) $(a * b) * c = a * (b * c)$
- (iii) (Identity) $\exists e \in G : e * a = a * e = a$
- (iv) (Inverse) $\exists a^{-1} \in G : a * a^{-1} = a^{-1} * a = e$

Additionally, a group is called **abelian** or **commutative** if it is

- (v) (Commutative) $a * b = b * a$

Examples. The **general linear group** $GL_n(\mathbb{R})$ consists of all matrices in $\mathbb{R}^{n \times n}$ with $\det \neq 0$. The **special linear group** $SL_n(\mathbb{R})$ additionally requires that $\det = 1$.

💡 Tip

The proof of (iii) and (iv) can be shortened if (v) is proven.

💡 Note

Defying conventional wisdom, **abelian** is usually not capitalized.

Definition 2.2 (Group Action)

A map $\cdot : G \times X \rightarrow X$ for a **(left) group action** if it satisfies:

- (i) (Identity) $e \cdot x = x$ for all $x \in X$.
- (ii) (Compatibility) $g \cdot (h \cdot x) = (gh) \cdot x$ for all $g, h \in G$ and $x \in X$.

2.2. SUBGROUPS

2.2.1. Definition

Let $(G, *)$ be a group.

Definition 2.3 (Subgroup)

A subset $H \subseteq G$ is a **subgroup** if it is itself a group under the operation inherited from G .

Theorem 2.1 (Subgroup Criterion)

A subset $H \subseteq G$ is a subgroup if and only if

- (i) $H \neq \emptyset$
- (ii) $a * b^{-1} \in H \quad \forall a, b \in H$

or

- (i) $H \neq \emptyset$ (nonempty)
- (ii) $|H| < \infty$ (finite)
- (iii) $a * b \in H \quad \forall a, b \in H$ (closed)

2.2.2. Cyclic

Definition 2.4 (Cyclic Subgroup)

For any $a \in G$, the set

$$\langle a \rangle = \{a^k \mid k \in \mathbb{Z}\}$$

forms a **cyclic subgroup** of G .

Theorem 2.2

The cyclic subgroup $\langle a \rangle \leq G$ is the smallest subgroup containing a .

Definition 2.5 (Generator)

An element $a \in G$ is a **generator** of G if $\langle a \rangle = G$.

Definition 2.6 (Cyclic Group)

We say G is a **cyclic group** if it contains a **generator** of itself.

Theorem 2.3

All subgroups of a cyclic group are cyclic.

Theorem 2.4

Every cyclic group is abelian.

Proof. Let g and γ be elements of a cyclic group $G = \langle a \rangle$. Then there are integers m and n such that $g = a^m$ and $y = a^n$, so

$$g\gamma = a^m a^n = a^{m+n} = a^{n+m} = a^n a^m = \gamma g.$$

2.2.3. Families

— Commutation

Let X be any nonempty subset of G .

Definition 2.7 (Centralizer)

The **centralizer** of X in G is the set

$$C_G(X) = \{g \in G \mid g x g^{-1} = x \text{ for all } x \in X\},$$

all elements in G that commute with all elements in X , fixing every element of X under conjugation.

💡 Remark

$$g x g^{-1} = x \iff g x = x g$$

💡 Note

If X is a singleton set $\{x\}$, we simply write $C_G(X)$.

Definition 2.8 (Center)

The **center** of G is the set

$$Z(G) = C_G(G) = \{g \in G \mid g \gamma g^{-1} = \gamma \text{ for all } \gamma \in G\},$$

all elements in G that commute with all elements in G , fixing every element of G under conjugation.

Definition 2.9 (Normalizer)

The **normalizer** of X in G is the set

$$N_G(X) = \{g \in G \mid g X g^{-1} = X\},$$

all elements in G that commute with X as a set, fixing X as a set under conjugation.

Theorem 2.5

$Z(G) \subseteq C_G(X) \subseteq N_G(X) \leq G$, with all equalities if G is abelian.

— Group Action

Let $* : G \times X \rightarrow X$ be a group action.

Definition 2.10 (Stabilizer)

The **stabilizer** of $x \in X$ is the set

$$G_x = \{g \in G \mid g * x = x\},$$

all elements in G that fix x under the group action $*$.

2.3. COSETS

2.3.1. Left and Right

Let H be a subgroup of a group G .

Definition 2.11 (Coset)

For any $g \in G$, the **left coset** of H in G is

$$gH = \{g * h \mid h \in H\}.$$

Similarly, the **right coset** is

$$Hg = \{h * g \mid h \in H\}.$$

Theorem 2.6

The left and right cosets respectively partition G .

Theorem 2.7

There are equally many left and right cosets of H in G .

2.3.2. Order

Definition 2.12 (Order)

The **order of a group** $|G|$ is the number of elements in G .

The **order of an element** $|g|$ is the smallest positive integer n such that $g^n = e$. If there is no such n , the order is infinite.

Definition 2.13 (Index)

The **index** of H in G is the number of left cosets, denoted $[G : H]$.

Theorem 2.8 (Lagrange's)

Let G be finite. Then the order of H divides the order of G .

Corollary

If G is of prime order, it

- (i) lacks proper nontrivial subgroups
- (ii) is generated by any element $g \neq e$
- (iii) is cyclic

Theorem 2.9 (Lagrange's, Generalized)

Regardless of whether G is finite or infinite,

$$|G| = [G : H] |H|.$$

Proof. G has $|G|$ elements and is partitioned into $[G : H]$ distinct left cosets, each of which has $|H|$ elements.

Theorem 2.10

Let $(g, h) \in G \times H$ be a tuple of groups. If they have finite orders $|g| = r < \infty$ and $|h| = s < \infty$, then $|(g, h)| = \text{lcm}(r, s)$.

Corollary

$$\mathbb{Z}_m \times \mathbb{Z}_n \cong \mathbb{Z}_{mn} \iff \text{gcd}(m, n) = 1$$

2.3.3. Conjugation

Definition 2.14 (Conjugation)

The **conjugation** by $a \in G$ is an **inner automorphism** of G defined $g \mapsto g a g^{-1}$.

Definition 2.15 (Normal Subgroup)

The following are equivalent for a subgroup H of G :

- (i) H is a **normal subgroup** of G , denoted $H \triangleleft G$.
- (ii) $ghg^{-1} \in H$ for all $g \in G$ and $h \in H$.
- (iii) $gHg^{-1} \subseteq H$ for all $g \in G$.
- (iv) $gHg^{-1} = H$ for all $g \in G$.

- (v) $gH = Hg$ (equal left coset and right coset) for all $g \in G$.
- (vi) The sets of left and right cosets coincide.

Examples. Trivially, G and $\{e\}$ are always normal subgroups.

Definition 2.16 (Simple Group)

A group is **simple** if it has no normal nontrivial proper subgroup.

Theorem 2.11

All subgroups of abelian groups are normal subgroups.

2.4. RINGS

2.4.1. Definition

Definition 2.17 (Ring)